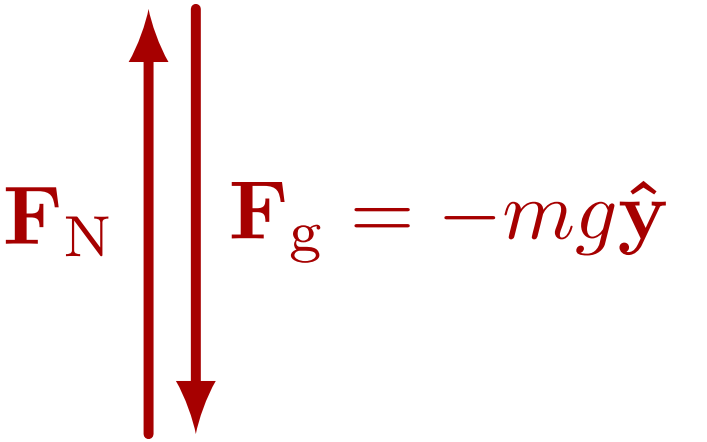
$\mathbf{F}_{\mathrm{N}}$
$\mathbf{F}_{\mathrm{g}} = -mg\hat{\mathbf{y}}$